Edward Silva

linkedin.com/in/edwardasilva | edwardasilva.com | eedwardasilva@gmail.com | (702) 720-7735

Experience

Embedded Software Engineer, SpaceX — Jul 2026 – Present
C, C++, Linux, Git — Bastrop, TX

- Write C and C++ software and Linux kernel drivers shipped on Starlink customer hardware
- Responsible engineer for the phased array software on 2 board revisions
- Brought up a new board revision from bare board to talking to satellites in 4 hours, down from 12+ hours
- Designed an algorithm that cut the run time of a phased array calculation by 11% across the deployed fleet
- Own the effective isotropic radiated power (EIRP) calculation system used by the RF and silicon teams

Software Engineering Intern, Kratos Defense — Jun – Aug 2025
C++, Python, SIMD, Git — Colorado Springs, CO

- Sped up legacy real-time C++ DSP code 1.6x with AVX-512 vectorization and refactoring, guided by Intel VTune
- Presented a faster FIR filter implementation and its performance trade-offs to the team for use on later projects
- Built a logging framework with configurable severity levels and error tracking for the real-time DSP software

Projects

Nonlinear State Estimation for the EMBR Damper, MATLAB, Simulink — Spring 2026

- Modeled the magnetorheological (MR) damper, coil, and sensors as a 4-state nonlinear system in Simulink
- Built extended (EKF) and unscented (UKF) Kalman filters to estimate the terrain force, which no sensor measures
- Verified every estimated state stayed inside its 3-sigma bound while the coil current stepped from 1 A to 2 A
- Showed the two filters agree within 0.4 N at the original parameters, where the fluid's nonlinear force is tiny
- Cut RMS force error from 624 N with the EKF to 401 N with a tuned UKF when the fluid's yield force was large
- Showed the EKF turns overconfident there, with only 58% of samples inside its 3-sigma bound

Autonomous Maze-Navigating Robot, Python — Fall 2025

- Co-built the OpenCV vision system for a 4-person maze robot that posted the second-fastest time in its cohort
- Estimated the distance and angle to each printed marker in real time with OpenCV ArUco detection
- Recalibrated the camera with more and sharper marker photos, and undistorted each marker before measuring it
- Split the 640x480 vision pipeline into 4 threads so the Arduino motor controller no longer waited on it
- Fixed an I2C packet mismatch with the Arduino by repacking 8-byte Python doubles as 4-byte little-endian floats
- Owned the team's Git repository and integration across all four members' subsystems

Skills

Skills: C++, Python, C, SIMD, MATLAB, Simulink, Git, Arduino

Education

Colorado School of Mines — May 2026
BS Electrical Engineering, Minor in Computer Science
Relevant Coursework: Signals and Systems I/II, Algorithms, Digital Signal Processing, Estimation Theory & Kalman Filtering, Modern Control Design, Embedded Systems, Computer Architecture, Software Engineering